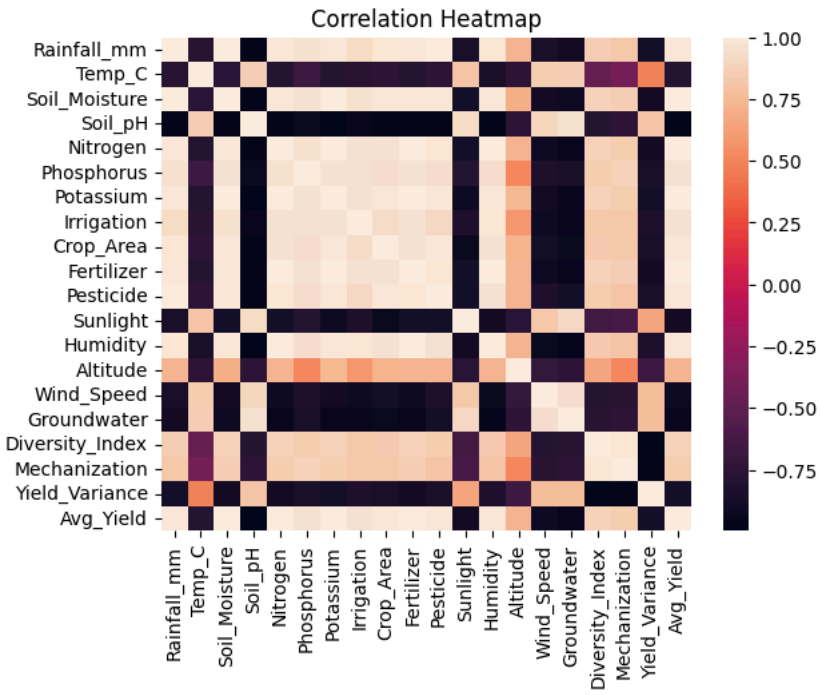
EDA ANALYSIS

```
#BASIC DATA INFORMATION
print(df.head())
print(df.tail())
print(df.info())

#SHAPE AND SIZE
print(df.shape)
print(df.size)

#NULL VALUES
print(df.isna().sum())
# Fill null values with the mean of their respective columns
df = df.fillna(df.mean())

# Verify that there are no more null values
print(df.isnull().sum().sum())
```

CORELATION ACROSS REGIONS

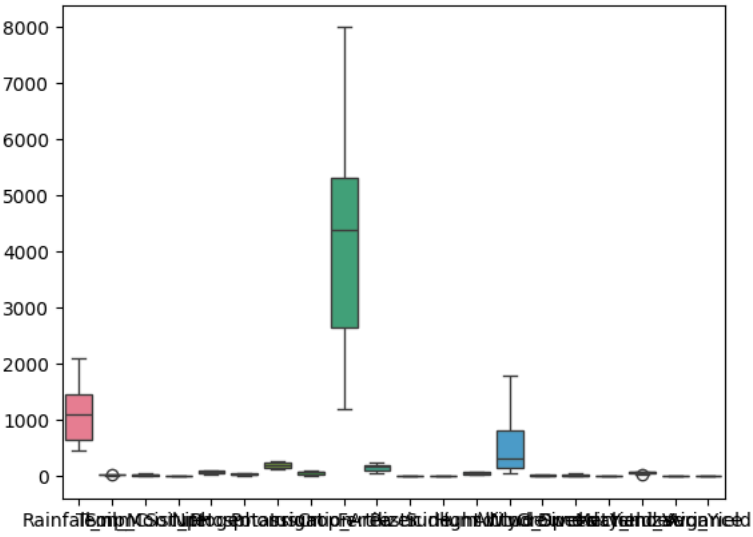
```
df.corr(numeric_only=True)
sns.heatmap(df.corr(numeric_only=True))
plt.title("Correlation Heatmap")
plt.show()
```



DATA NORMALIZATION [MIN-MAX SCALER]

```
scaler = StandardScaler()
scaled_array = scaler.fit_transform(df)
df_normalized = pd.DataFrame(scaled_array, columns=df.columns)
print(df_normalized)
```

```
import seaborn as sns
sns.boxplot(df)
plt.show()
```

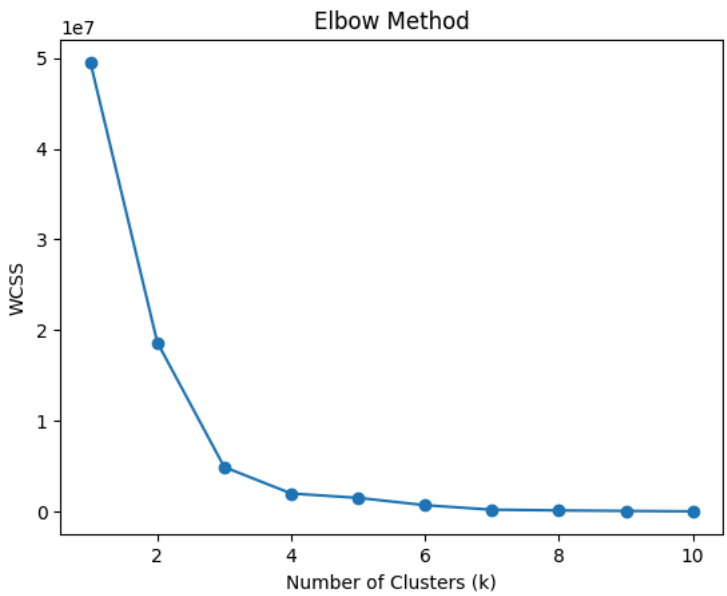


FORWARD/BACKWARD FEATURE SELECTION

```
x = df.iloc[:,:-1]
y = df.iloc[:,-1]
model = LinearRegression()
fs = SequentialFeatureSelector(model, n_features_to_select=3, direction='backward')
fs.fit(x, y)
```

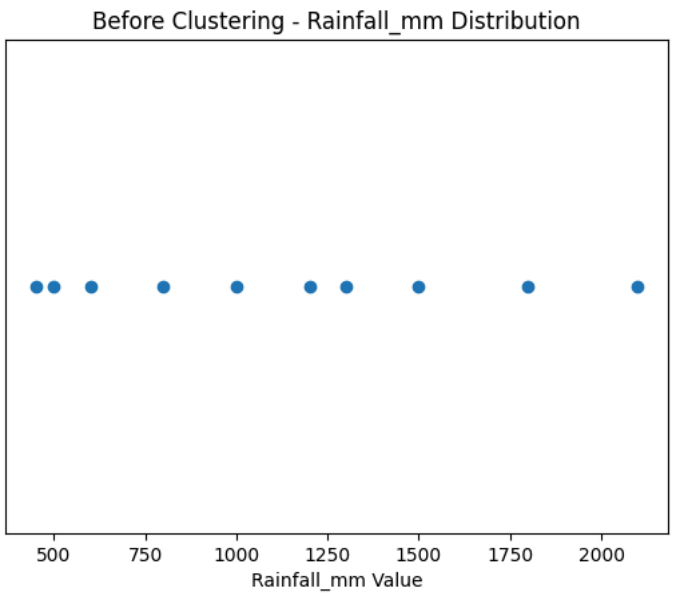
K-CLUSTERING

```
X = df.values
wcss = []

for k in range(1, 11):
    kmeans = KMeans(n_clusters=k, random_state=42)
    kmeans.fit(X)
    wcss.append(kmeans.inertia_)

plt.plot(range(1, 11), wcss, marker='o')
plt.xlabel("Number of Clusters (k)")
plt.ylabel("WCSS")
plt.title("Elbow Method")
plt.show()
```

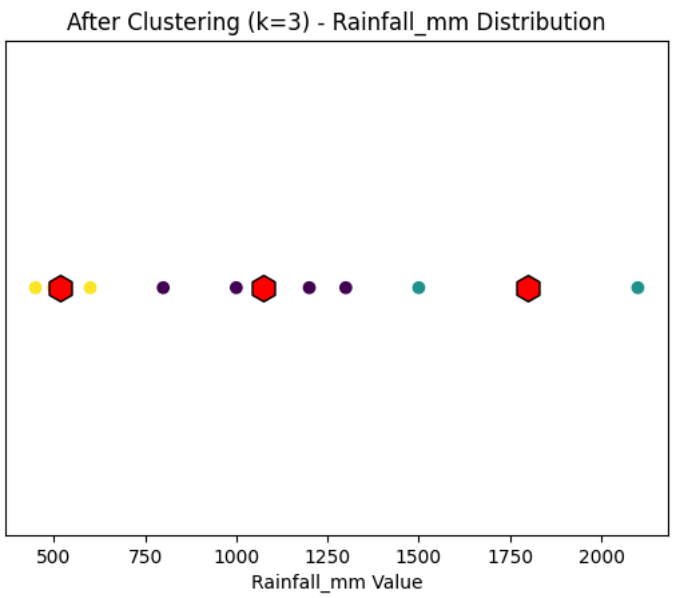



```
# before clustering
plt.figure()
#to plot 'Rainfall_mm'
plt.scatter(df['Rainfall_mm'], [0]*len(df))
plt.title("Before Clustering - Rainfall_mm Distribution")
```

```
plt.xlabel("Rainfall_mm Value")
plt.yticks([])
plt.show()
```




```
# After clustering
plt.figure()

optimal_k = 3
kmeans_final = KMeans(n_clusters=optimal_k, random_state=42, n_init='auto')
kmeans_final.fit(X)
df['Cluster'] = kmeans_final.labels_
#cluster centroids
centroids = kmeans_final.cluster_centers_
plt.scatter(df["Rainfall_mm"], [0]*len(df), c=df["Cluster"])
plt.scatter(centroids[:, 0], [0]*len(centroids), marker='h', s=200, color='red', edgecolors='black')

plt.title(f"After Clustering (k={optimal_k}) - Rainfall_mm Distribution")
plt.xlabel("Rainfall_mm Value")
plt.yticks([])
plt.show()

print("\nClustered Data:")
print(df)
wcss
```

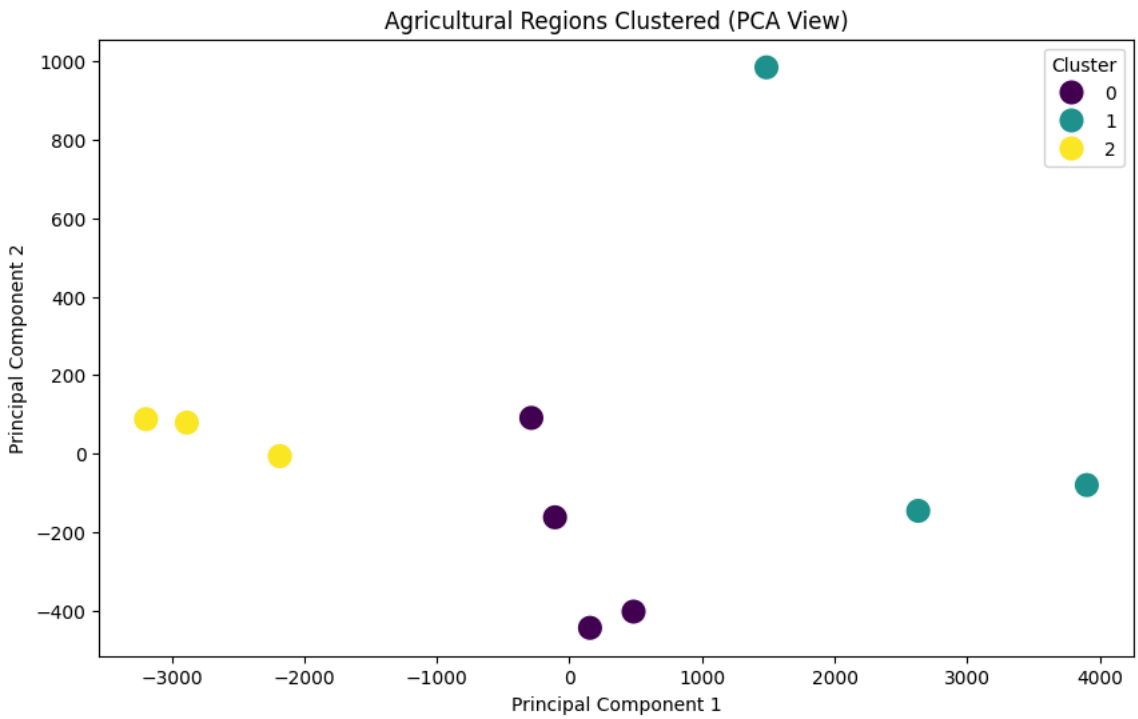


SILHOUETTE SCORE

```
silhouette_avg = silhouette_score(X, df['Cluster'])
print(f"The average silhouette score is: {silhouette_avg}")
```

PCA FOR 2D VISUALIZATION

```
pca = PCA(n_components=2)
pca_data = pca.fit_transform(X)

plt.figure(figsize=(10, 6))
sns.scatterplot(x=pca_data[:, 0], y=pca_data[:, 1], hue=df['Cluster'], palette='viridis', s=200)
plt.title('Agricultural Regions Clustered (PCA View)')
plt.xlabel('Principal Component 1')
plt.ylabel('Principal Component 2')
plt.show()
```



MEAN CHARACTERISTIC OF EACH CLUSTER

```
print("\n--- Cluster Profiles (Mean Values) ---")
print(df.groupby('Cluster').mean(numeric_only=True))
```

PREDICTED CLUSTER

```
predicted_clusters = kmeans_final.predict(X)
df['Predicted_Cluster_Label'] = predicted_clusters
print("DataFrame with Predicted Cluster Labels:")
print(df[['Rainfall_mm', 'Avg_Yield', 'Cluster', 'Predicted_Cluster_Label']].head())
```